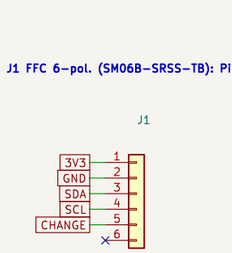
U1 AT42QT2120 (VQFN-20, Comms-Modus): MODE an GND, I2C 0x1C, KEY0-2 = Wheel, KEY3 = Mitteltaste, KEY4 = Guard

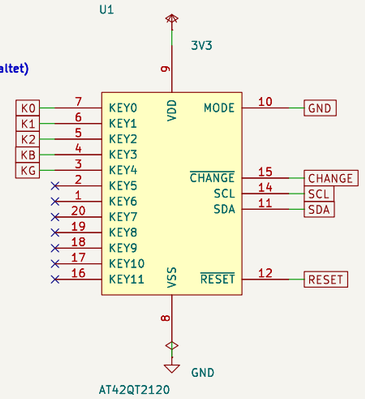
Wie Tangara: kein Cs noetig (QT2120-Datenblatt: no external Cs required), RESET ueber 10k an 3V3, Exposed Pad ohne Anschluss

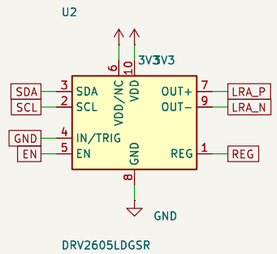
Pins 1, 2, 16-20 (KEY5-KEY11) unbenutzt. CHANGE ist open drain: Pull-up auf der MCU-Seite (wie bei Tangara: interner Pull-up der MCU).

U2 DRV2605L (VSSOP-10): I2C 0x5A, LRA-Modus, IN/TRIG an GND, EN ueber 10k an 3V3 (wie Tangara)

J1 FFC 6-pol. (SM06B-SRSS-TB): Pin 6 = BTN (Reserve, nicht beschaltet)



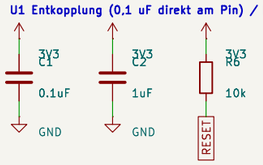




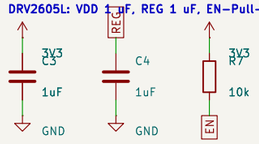
Loetpads LRA-Litzen (Rueckseite)



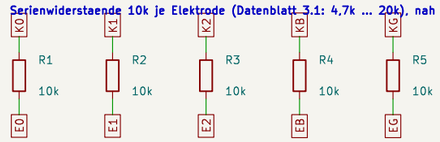
U1 Entkopplung (0,1 uF direkt am Pin) / RESET-Pull-up



DRV2605L: VDD 1 uF, REG 1 uF, EN-Pull-up (wie Tangara)



Serienwiderstaende 10k je Elektrode (Datenblatt 3.1: 4,7k ... 20k), nah am Chip



Kupferflaechen vorn (Lackschicht geschlossen): Wheel (3 Elektroden), Mitteltaste, Guard (3 Boegen, ein Netz)



I2C-Pull-ups 4,7k: DNP (das Waveshare-Board hat 2,2k). Nur bestuecken, wenn das Modul allein getestet wird.



Schnittstelle laut TEILE.md, Abschnitt Klickrad-Modul; FFC 6-pol., Pin 5 = CHANGE, Pin 6 = Reserve
AT42QT2120 (0x1C, Wheel) + DRV2605L (0x5A, LRA), Beschaltung nach Tangara-Faceplate (CERN-OHL

Nano-Player (Hobbyprojekt)

Sheet: /
File: klickrad.kicad_sch

Title: Klickrad-Modul v2

Size: A3	Date: 2026-10-04	**Rev: 2**
KiCad E.D.A. 9.0.9		Id: 1/1